



- **Classical communication complexity problems (10.3.1)**

多个参与者各自持有一部分输入，为了共同计算某个函数或完成某个任务，至少需要交换多少信息；CC, RCC, RCC(pub), DCC, IC(mu), IC;

- **Deterministic communication: lower bounds and structure (10.3.2)**

rectangle, f-constant, fooling set, rectangle measure;

- **Randomization, conditional information complexity and direct sum (10.3.3)**

primitives $\rightarrow$ global using planted solution and fooling distribution;

- **Direct sum communication bounds (10.4.1)**

把统计估计问题放到一个信息受限的通信框架中研究；建立高维估计问题的 direct sum 下界，把 d-维问题的误差下界归约到 d 个一维检验问题的误差下界之和

- **Communication data processing (10.4.2)**

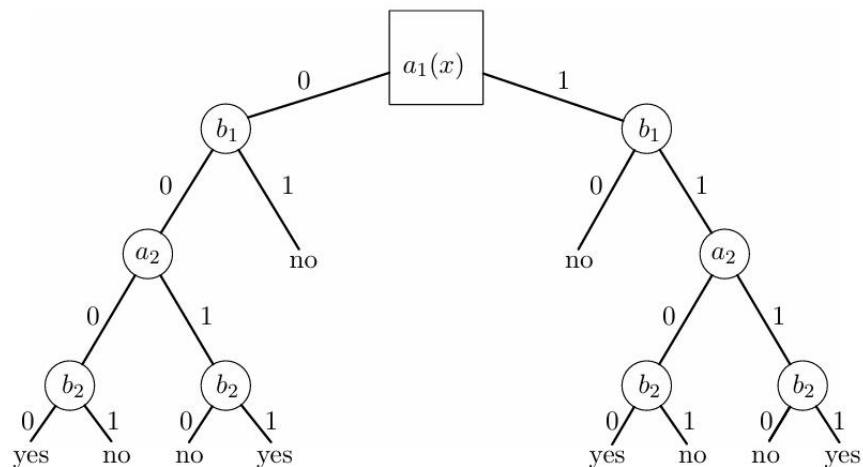
先控制单个二元坐标通信在 transcript 下的 Hellinger 可区分性，再通过 Assouad 方法累加到 d 个坐标上





Classical communication complexity problems

The most basic problems in communication complexity ask a simple question: **two entities** (always named Alice and Bob) have **inputs x, y** and wish to jointly **compute a function $f(x, y)$** . How many bits—or other messages—must Alice and Bob communicate to compute this value? Less abstractly, Alice and Bob have input domains $\mathcal{X}$ and $\mathcal{Y}$ (often, these are $\{0, 1\}^n$), and Alice receives a vector $x \in \mathcal{X}$ and Bob $y \in \mathcal{Y}$, each unknown to the other, and they jointly exchange messages until they can successfully evaluate $f(x, y)$. To abstract away any details of the computational model, we assume each has infinite computational power, which allows a focus on communication. To formulate this as communication, we consider a **protocol Π** , which specifies the messages that each of Alice and Bob send to one another. We view this as a series of rounds, where at each round, the protocol allows one $\{0, 1\}$ -valued bit to be sent and determines who sends this bit, and, at termination time, can compute $f(x, y)$ based on the communicated message. Then the **communication cost** of Π is the **maximum number of messages sent to (correctly) compute f over all inputs x, y** .



inputs x, y ;

compute $f(x, y)$ by communication protocol Π





Classical communication complexity problems

Definition 10.3. A protocol Π over a domain $\mathcal{X} \times \mathcal{Y}$ with output space $\mathcal{Z}$ is a **binary tree**, where each internal node v is labeled with a **mapping** $a_v : \mathcal{X} \rightarrow \{0, 1\}$ or $b_v : \mathcal{Y} \rightarrow \{0, 1\}$ and each leaf is labeled with a **value** $z \in \mathcal{Z}$.

In classical communication complexity, the main questions center around the *communication complexity* of $f : \mathcal{X} \times \mathcal{Y} \rightarrow \mathcal{Z}$, which is the **length of the shortest protocol that computes f correctly on all inputs**: letting $\Pi_{\text{out}}(x, y)$ denote the final output of the protocol Π on inputs (x, y) , this is

$$\text{CC}(f) := \inf \{ \text{depth}(\Pi) \mid \Pi_{\text{out}}(x, y) = f(x, y) \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y} \}.$$

In many cases, it is useful to allow **randomized communication protocols**, which tolerate some **probability of error**; in this case, we let Alice and Bob each have access to (an arbitrary amount) of randomness, which we can identify without loss of generality with uniform random variables $U_a, U_b \stackrel{\text{iid}}{\sim} \text{Uniform}[0, 1]$. The nodes a_v and b_v in Definition 10.3 are then mappings $a_v : \mathcal{X} \times [0, 1] \rightarrow \{0, 1\}$ and $b_v : \mathcal{Y} \times [0, 1] \rightarrow \{0, 1\}$, and they calculate $a_v(\cdot, U_a)$ and $b_v(\cdot, U_b)$, respectively. Abusing notation slightly by leaving this randomness implicit, the *randomized communication complexity* for an accuracy δ is then the length of the shortest randomized protocol that calculates $f(x, y)$ correctly with probability at least $1 - \delta$, that is,

randomize and tolerate
some probability of error

$$\text{RCC}_\delta(f) \leq O(1) \log \frac{1}{\delta} \cdot \text{RCC}_{1/3}(f),$$

$$\text{RCC}_\delta(f) := \inf \{ \text{depth}(\Pi) \mid \mathbb{P}(\Pi_{\text{out}}(x, y) \neq f(x, y)) \leq \delta \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y} \}. \quad (10.3.1)$$





Classical communication complexity problems

There are variants of randomized complexity that allow **public randomness** rather than private randomness, which can yield simpler algorithms and somewhat reduced complexity, but this improvement is limited, as Alice and Bob can more or less simulate public randomness (see Exercise 10.7). Letting $\mathfrak{P}_{\text{pub}}$ be the collection of protocols in which both Alice and Bob have access to a **shared random variable** $U \sim \text{Uniform}[0, 1]$, we make the obvious extension

$$\text{RCC}_{\delta}^{\text{pub}}(f) := \inf_{\Pi \in \mathfrak{P}_{\text{pub}}} \{ \text{depth}(\Pi) \mid \mathbb{P}(\Pi_{\text{out}}(x, y, U) \neq f(x, y)) \leq \delta \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y} \}.$$

compare $\text{RCC}_{\delta}(f) := \inf \{ \text{depth}(\Pi) \mid \mathbb{P}(\Pi_{\text{out}}(x, y) \neq f(x, y)) \leq \delta \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y} \}.$ (10.3.1)

Finally, we have notions of communication complexity admitting the most powerful algorithms and so providing the strongest lower bounds. The first of these is **distributional communication complexity**, which for a probability measure μ on inputs $\mathcal{X} \times \mathcal{Y}$ is the depth of the shortest protocol that succeeds with a given μ -probability:

$$\text{DCC}_{\delta}^{\mu}(f) := \inf \{ \text{depth}(\Pi) \mid \mu(\Pi_{\text{out}}(X, Y) \neq f(X, Y)) \leq \delta \}, \quad (10.3.2)$$

where the infimum is taken over **deterministic** protocols. As in the minimax risk, once we reduce to distributional complexity, using deterministic protocols is no loss, as if for a random variable U the randomized protocol satisfies $\mu(\Pi_{\text{out}}(X, Y, U) \neq f(X, Y)) \leq \delta$, then certainly $\inf_u \mu(\Pi_{\text{out}}(X, Y, u) \neq f(X, Y)) \leq \delta$. The final notion we consider is the *information complexity*.

mu 描述不同 (x,y) 出现的概率

infimum only on deterministic protocols, since stochastic 可以视为 deterministic 的平均





Classical communication complexity problems

$\inf_u \mu(\Pi_{\text{out}}(X, Y, u) \neq f(X, Y)) \leq \delta$. The final notion we consider is the *information complexity*. In this case, we require again that for each input pair x, y , the (potentially randomized) protocol $\Pi(x, y)$ still compute $f(x, y)$ correctly with probability at least $1 - \delta$, but instead of measuring the depth of the tree, we let X, Y be drawn randomly from some distribution and measure the mutual information $I_2(X, Y; \Pi(X, Y))$. (We use base-2 logarithms to reflect bit communication.) In this case, we define

$$IC_\delta(f) := \sup_{\Pi} \inf \{I_2(X, Y; \Pi(X, Y)) \mid \mathbb{P}(\Pi_{\text{out}}(x, y) \neq f(x, y)) \leq \delta \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y}\}, \quad (10.3.3)$$

$$IC_\delta^\mu(f) := \inf_{\Pi} \{I_2(X, Y; \Pi(X, Y)) \mid \mu(\Pi_{\text{out}}(X, Y) \neq f(X, Y)) \leq \delta\},$$

Proposition 10.3.1. For any function f , $\delta \in (0, 1)$, and probability measure μ on $\mathcal{X} \times \mathcal{Y}$,

$$CC(f) \geq RCC_\delta(f) \geq RCC_\delta^{\text{pub}}(f) \geq DCC_\delta^\mu(f) \geq IC_\delta^\mu(f)$$

and

$$RCC_\delta(f) \geq IC_\delta(f).$$

$I(X, Y; \Pi(X, Y))$ 表示通信 transcript 中包含多少关于完整输入 (X, Y) 的信息

最坏输入分布下的最小信息代价，
sup 对 μ 取





Classical communication complexity problems

Proposition 10.3.1. For any function f , $\delta \in (0, 1)$, and probability measure μ on $\mathcal{X} \times \mathcal{Y}$,

$$\text{CC}(f) \geq \text{RCC}_\delta(f) \geq \text{RCC}_\delta^{\text{pub}}(f) \geq \text{DCC}_\delta^\mu(f) \geq \text{IC}_\delta^\mu(f)$$

and

$$\text{RCC}_\delta(f) \geq \text{IC}_\delta(f).$$

Proof The first two inequalities are immediate. By Theorem 2.4.3, we have

$$\text{depth}(\Pi) \geq H_2(\Pi) \geq H_2(\Pi) - H_2(\Pi \mid X, Y) = I_2(X, Y; \Pi(X, Y)),$$

and so for all $\delta \in (0, \frac{1}{2})$ we have both

$$\text{RCC}_\delta(f) \geq \text{IC}_\delta(f) \quad \text{and} \quad \text{DCC}_\delta^\mu(f) \geq \text{IC}_\delta^\mu(f).$$

All that remains is to demonstrate $\text{RCC}_\delta^{\text{pub}}(f) \geq \text{DCC}_\delta^\mu(f)$. For this, let Π be any protocol with public randomness U such that $\mathbb{P}(\Pi_{\text{out}}(x, y, U) \neq f(x, y)) \leq \delta$ for all x, y . Then by taking an expectation over $(X, Y) \sim \mu$, we obtain

$$\delta \geq \mathbb{E}_\mu [\mathbb{P}(\Pi_{\text{out}}(X, Y, U) \neq f(X, Y) \mid X, Y)] \geq \inf_u \mu(\Pi_{\text{out}}(X, Y, u) \neq f(X, Y)),$$

that is, there must be at least some u achieving the average error of Π , and the protocol Π is deterministic given u . So any protocol Π using public randomness to achieve probability of error δ can be modified into a deterministic protocol $\Pi(\cdot, \cdot, u)$ that achieves μ -probability of error δ .¹ $\square$

$I(X, Y; \Pi(X, Y))$ 表示通信 transcript 中包含多少关于完整输入 (X, Y) 的信息

最坏输入分布下的最小信息代价，
sup 对 μ 取

$H_2(\Pi)$ denotes Shannon entropy





Deterministic communication: lower bounds and structure

A set $R \subset \mathcal{X} \times \mathcal{Y}$ is a **rectangle** if it has the form $R = A \times B$ for some $A \subset \mathcal{X}$ and $B \subset \mathcal{Y}$. Equivalently, R is a rectangle if $(x_0, y_0) \in R$ and $(x_1, y_1) \in R$ imply that $(x_0, y_1) \in R$. Rectangular sets provide a key to understanding communication complexity:

Proposition 10.3.2. *Let v be a node in a deterministic protocol Π and R_v be those pairs (x, y) reaching node v . Then R_v is a rectangle.*

The structure of rectangles for correct protocols thus determines the communication complexity of a function f . For a set $R \subset \mathcal{X} \times \mathcal{Y}$, we say R is **f -constant** if $f(x, y) = f(x', y')$ for all $(x, y) \in R$ and $(x', y') \in R$. Thus, any correct protocol Π partitions $\mathcal{X} \times \mathcal{Y}$ into a collection of f -constant rectangles, where we identify the rectangles with the leaves l of the protocol tree. In particular, Proposition 10.3.2 implies the following corollary.

Corollary 10.3.3. *Let N be the size of the minimal partition of $\mathcal{X} \times \mathcal{Y}$ into f -constant rectangles. Then $\text{CC}(f) \geq \log_2 N$.*

Definition 10.4 (Fooling sets). *A set $S \subset \mathcal{X} \times \mathcal{Y}$ is a fooling set for f if for any two pairs $(x_0, y_0) \in S$ and $(x_1, y_1) \in S$ satisfying $f(x_0, y_0) = f(x_1, y_1)$, at least one of the inequalities $f(x_0, y_1) \neq f(x_0, y_0)$ or $f(x_1, y_0) \neq f(x_0, y_0)$ holds.*

With this definition, the next corollary is almost immediate.

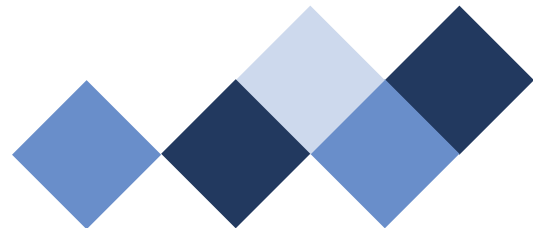
Corollary 10.3.4. *Let f have a fooling set S of size N . Then $\text{CC}(f) \geq \log_2 N$.*

An extension of the fooling set idea is the **rectangle measure** method, which proves that (for some probability measure P) the “size” of f -constant rectangles is small. By judicious choice of the probability, we can then demonstrate lower bounds.

Proposition 10.3.5. *Let P be a probability distribution on $\mathcal{X} \times \mathcal{Y}$. If all f -constant rectangles R have probability at most $P(R) \leq \delta$, Then $\text{CC}(f) \geq \log_2 \frac{1}{\delta}$.*

The structures are as follows:

- rectangle
- f -constant rectangles
- fooling sets
- rectangle measure





Example 10.3.6 (Equality): Consider the problem of testing equality of two n -bit strings $x, y \in \{0, 1\}^n$, letting $f = \text{EQ}$ be $f(x, y) = 1$ if $x = y$ and 0 otherwise. Define the set $S = \{(x, x) \mid x \in \{0, 1\}^n\}$, which has cardinality 2^n , and satisfies $f(x, x) = 1$ for all $(x, x) \in S$.

two example, 分別用到 fooling set
和 rectangle measure

That S is a fooling set is immediate: for any (x, x) and $(x', x') \in S$, if $x \neq x'$, then certainly $(x, x') \notin S$. So

$$n \leq \text{CC}(\text{EQ}) \leq n + 1,$$

Example 10.3.7 (Inner products on $\mathbb{F}_2$): Consider computing the inner product $\text{IP}_2(x, y) = \langle x, y \rangle \bmod 2$ for n -bit strings $x, y \in \{0, 1\}^n$, where addition is performed modulo 2. Rather than constructing a fooling set directly, we use Proposition 10.3.5 and let P be the uniform distribution on $\{0, 1\}^n \times \{0, 1\}^n$. Let $R = A \times B$ be a rectangle with $\langle x, y \rangle = 0$ for all $x \in A$ and $y \in B$. The linearity of the inner product guarantees that $\langle x, y \rangle = 0$ for all $x \in \text{span}(A)$ and $y \in \text{span}(B)$, the (linear) spans of A and B in $\mathbb{F}_2^n$, respectively. Now recognize that $\text{span}(A), \text{span}(B) \subset \mathbb{F}_2^n$ are orthogonal subspaces of $\mathbb{F}_2^n$, and so their dimensions $d_0 = \dim(\text{span}(A))$ and $d_1 = \dim(\text{span}(B))$ satisfy $d_0 + d_1 \leq n$.

Noting that if $d_0 = \dim(A)$ then $|A| \leq 2^{d_0}$ in $\mathbb{F}_2^n$, we thus obtain $|R| \leq |A| \cdot |B| \leq 2^n$, which (under the uniform measure P) satisfies

$$P(R) \leq \frac{2^n}{2^{2n}} = 2^{-n}.$$

By Proposition 10.3.5, we thus have

$$n \leq \text{CC}(\text{IP}_2) \leq n + 1,$$





Example 10.3.8 (Equality with randomization): Let $x, y \in \{0, 1\}^n$ and p be a prime number satisfying $n^2 \leq p \leq 2n^2$ (the Prime Number Theorem guarantees the existence of such a p). Let Alice choose a uniformly random number $U \in \{0, \dots, p-1\}$ and compute the polynomial

$$a(U) = x_1 + x_2U + x_3U^2 + \dots + x_nU^{n-1} \pmod{p}.$$

Then Alice may communicate both U and $a(U)$ to Bob, which requires at most $2 \log_2 p \leq 4 \log_2 n + 2 \log 2$ bits. Then Bob checks whether

$$b(U) = y_1 + y_2U + y_3U^2 + \dots + y_nU^{n-1} \pmod{p}$$

satisfies $b(U) = a(U)$. If so, Bob outputs “Yes” (equality), and otherwise, Bob outputs “No.” This protocol satisfies $\text{depth}(\Pi) \leq 4 \log_2 n + 1$. Moreover, if $x = y$, it is always correct, while if $x \neq y$, then the protocol is incorrect only if $a(U) = b(U)$, that is, U is a root of the polynomial

$$p(u) = \sum_{i=1}^n (x_i - y_i)u^{i-1}.$$

But this is a non-zero degree $n-1$ polynomial, which has at most $n-1$ roots (on the field $\mathbb{F}_p$; see Appendix A.1 for a brief review of polynomials). Thus for $x \neq y$ we have

$$\mathbb{P}(\Pi(x, y) \text{ fails}) = \mathbb{P}(a(U) = b(U)) \leq \frac{n-1}{p} < \frac{1}{n},$$

and so $\text{RCC}_{1/n}(\text{EQ}) \leq O(1) \log n$, exponentially improving over the deterministic complexity.

随机化大大缩小所需的信息量

证明也不能使用之前的 rectangle 等简单工具，作者将引入一些更适合信息论分析的方法

证明 randomized communication complexity 的经典工具是 discrepancy and randomized inputs，但这里的重点是 information-theoretic techniques

- direct sum inequalities
- lower bound for fundamental primitive
- primitive + direct sum = global lower bound





Direct sum bounds and decomposition

To show direct sum inequalities, we demonstrate that computing some function on n inputs requires roughly n times the communication of single-input computation. In general, we consider functions f of the form

$$f(x_1^n, y_1^n) = g(h(x_1, y_1), h(x_2, y_2), \dots, h(x_n, y_n)), \quad (10.3.4)$$

where g is the global function of the n primitives h , calling such functions decomposable with primitive h . Many problems have the structure (10.3.4); focusing on the case that the inputs $x, y \in \{0, 1\}^n$ and $f(x, y) \in \{0, 1\}$, we have the following three immediate examples.

Example 10.3.9 (Decomposition of equality): The equality function $f(x, y) = 1$ if $x \neq y$ and $f(x, y) = 0$ otherwise satisfies the decomposition (10.3.4), where $h(x_i, y_i) = \mathbf{1}\{x_i \neq y_i\}$ and g is the OR function $g(z) = \mathbf{1}\{\langle \mathbf{1}, z \rangle > 0\}$, which is 1 if any of $z_1, \dots, z_n$ is non-zero, and 0 otherwise. $\diamond$

Example 10.3.10 (Decomposition of inner product): The inner product in $\mathbb{F}_2$, $f(x, y) = \langle x, y \rangle \bmod 2$, where $h(x_i, y_i) = x_i y_i$, and $g(z) = \langle \mathbf{1}, z \rangle \bmod 2$ satisfies $g(z) = 0$ if $\sum_{i=1}^n z_i$ is even and $g(z) = 1$ otherwise. $\diamond$

Example 10.3.11 (Decomposition of disjointness): The set disjointness function $f(x, y) = \text{DISJ}(x, y) := \mathbf{1}\{\langle x, y \rangle > 0\}$ arises when x, y are characteristic vectors of subsets A, B of $[n]$, that is, $x_i = \mathbf{1}\{i \in A\}$ and $y_i = \mathbf{1}\{i \in B\}$. Then $f(x, y) = \mathbf{1}\{A \cap B \neq \emptyset\}$ corresponds to g being the OR $g(z) = \mathbf{1}\{\langle \mathbf{1}, z \rangle > 0\}$ and h the AND function $h(x_i, y_i) = x_i y_i$. $\diamond$

give f a structure





Direct sum bounds and decomposition

Definition 10.5. Let $f : \mathcal{X}^n \times \mathcal{Y}^n \rightarrow \{0, 1\}$ have the decomposition (10.3.4), where the primitive h is $\{0, 1\}$ -valued. The pair $(x, y) \in \mathcal{X}^n \times \mathcal{Y}^n$ admits a **planted solution** if for each $i \in \{1, \dots, n\}$, all x'_i, y'_i , and vectors

$$x' = (x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n) \text{ and } y' = (y_1, \dots, y_{i-1}, y'_i, y_{i+1}, \dots, y_n),$$

we have $f(x', y') = h(x'_i, y'_i)$.

Definition 10.6. A distribution μ on $(x, y) \in \mathcal{X} \times \mathcal{Y}$ is a **fooling distribution** if all (x_1^n, y_1^n) in the support of the product μ^n admit planted solutions (Definition 10.5).

把一个 n 维大问题局部嵌入成一个一维小问题

1. 假设有一个协议 Π 可以用很少信息解决大函数 f ;
2. 从 fooling distribution 里抽一个 planted background (X^n, Y^n) ;
3. 随机选一个坐标 i , 把这个坐标替换成新的 primitive 输入 (x'_i, y'_i) ;
4. 因为 background 是 planted solution, 所以此时:

$$f(x', y') = h(x'_i, y'_i).$$

5. 所以用解决 f 的协议 Π , 就可以解决 primitive h ;
6. 如果已知 primitive h 本身需要至少某个信息量, 那么大函数 f 的协议不可能信息量太小。

这就是 direct sum 的基本逻辑:

solving f on n coordinates $\implies$ solving one embedded h .

why choose fooling distribution?





Direct sum bounds and decomposition

plexity. If $(X, Y) \sim \mu$, there is always a random variable V such that $X \perp\!\!\!\perp Y \mid V$, that is, X and Y are **conditionally independent** given V (trivially, we can take $V = X$). Thus, for function $h : \mathcal{X} \times \mathcal{Y} \rightarrow \{0, 1\}$, we define the **conditional information complexity**

$$\text{CIC}_\delta^\mu(h) := \inf_{\Pi} \sup_V \{I(X, Y; \Pi(X, Y) \mid V) \text{ s.t. } \mathbb{P}(\Pi_{\text{out}}(x, y) \neq h(x, y)) \leq \delta \text{ for all } x \in \mathcal{X}, y \in \mathcal{Y}\},$$

Theorem 10.3.13. *Let μ be a fooling distribution $\mathcal{X} \times \mathcal{Y}$ for a function f with primitive h . Then*

$$\text{IC}_\delta(f) \geq n \cdot \text{CIC}_\delta^\mu(h).$$

Proof Let $V = V_1^n \in \mathcal{V}^n$ be any random vector with i.i.d. entries making (X_i, Y_i) conditionally independent given V_i . Then for any protocol Π , we have

$$\begin{aligned} I(X_1^n, Y_1^n; \Pi) &= H(\Pi) - H(\Pi \mid X_1^n, Y_1^n) \\ &= H(\Pi) - H(\Pi \mid X_1^n, Y_1^n, V) \geq H(\Pi \mid V) - H(\Pi \mid X_1^n, Y_1^n, V) = I(X_1^n, Y_1^n; \Pi \mid V) \end{aligned}$$

because we have the Markov chain $V \rightarrow (X_1^n, Y_1^n) \rightarrow \Pi$. Using the chain rule for mutual information, where we recognize that X_1^n and Y_1^n are independent given V , we have

$$\begin{aligned} I(X_1^n, Y_1^n; \Pi \mid V) &= \sum_{i=1}^n I(X_i, Y_i; \Pi \mid V, X_1^{i-1}, Y_1^{i-1}) \\ &= \sum_{i=1}^n H(X_i, Y_i \mid V, X_1^{i-1}, Y_1^{i-1}) - H(X_i, Y_i \mid V, \Pi, X_1^{i-1}, Y_1^{i-1}) \\ &\geq \sum_{i=1}^n H(X_i, Y_i \mid V) - H(X_i, Y_i \mid V, \Pi) = \sum_{i=1}^n I(X_i, Y_i; \Pi \mid V) \end{aligned} \quad (10.3.5)$$

conditionally independent

因为 X 和 Y 可能本来相关，不条件化， $I(X, Y; \Pi)$ 就会混杂两类互信息；

内层 **sup**: 无论如何解释 X, Y 之间的相关性，只要条件化后 X, Y 独立，协议仍然必须泄露多少额外信息；

外层 **inf**: 希望泄露信息数最少；

协议 Π 从整个输入中获得的信息，至少包含它分别从每个坐标 (X_i, Y_i) 中获得的信息的总和





Direct sum bounds and decomposition

We come to the key reduction from the global protocol Π to one solving individual primitives. On inputs $(x, y) \in \mathcal{X} \times \mathcal{Y}$, define the simulated protocol $\Pi_{i,v}(x, y)$ so that given the vector $v_{\setminus i} = (v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n) \in \mathcal{V}^{n-1}$, Alice and Bob independently generate $(X_j^*, Y_j^*) \stackrel{\text{iid}}{\sim} \mu(\cdot \mid V_j = v_j)$ for $j \neq i$, which is possible because of the assumed conditional independence given V , yielding $X_{\setminus i}^* \in \mathcal{X}^{n-1}$ and $Y_{\setminus i}^* \in \mathcal{Y}^{n-1}$, respectively. They then execute the protocol $\Pi((X_{\setminus i}^*, x), (Y_{\setminus i}^*, y))$ (where we substitute x and y into input position i for each). We claim two key consequences of this simulation: first, $\Pi_{i,v}$ is a δ -error protocol for the primitive h , and second, we have the distributional equality

$$(X_i, Y_i, V_i, \Pi_{i,v}(X_i, Y_i)) \stackrel{\text{dist}}{=} (X_i, Y_i, V_i, \Pi(X_1^n, Y_1^n)) \mid V_{\setminus i} = v_{\setminus i}, \quad (10.3.6)$$

that is, the joint over the simulated protocol is equal to that over the original protocol Π conditional on $V_{\setminus i} = v_{\setminus i}$. The claim (10.3.6) is essentially definitional. To see the first, that $\Pi_{i,v}$ is a δ -error

- 第 i 个坐标放入真正输入：

$$(X_i, Y_i) = (x, y).$$

- 其他坐标 $j \neq i$ 从 fooling distribution 中随机生成：

$$(X_j^*, Y_j^*) \sim \mu(\cdot \mid V_j = v_j).$$

于是得到完整输入：

$$((X_{\setminus i}^*, x), (Y_{\setminus i}^*, y)).$$

然后运行原来的协议：

$$\Pi((X_{\setminus i}^*, x), (Y_{\setminus i}^*, y)).$$

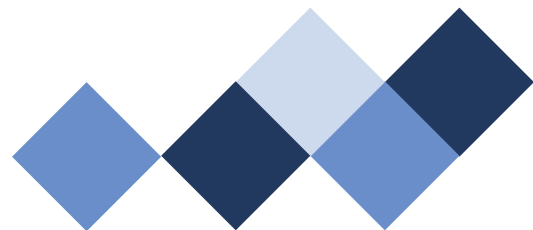
这个新协议就叫

$$\Pi_{i,v}(x, y).$$

因此对任意 primitive 输入 (x, y) ，都有

$$f((X_{\setminus i}^*, x), (Y_{\setminus i}^*, y)) = h(x, y).$$

所以，如果原协议 Π 能以错误率 δ 解决大函数 f ，那么模拟协议 $\Pi_{i,v}$ 也能以错误率 δ 解决 primitive h 。





Direct sum bounds and decomposition

$$(X_i, Y_i, V_i, \Pi_{i,v}(X_i, Y_i)) \stackrel{d}{=} (X_i, Y_i, V_i, \Pi(X_1^n, Y_1^n)) \mid V_i = v_i.$$

意思是：

在固定其他坐标的辅助变量 $V_{\setminus i} = v_{\setminus i}$ 后，模拟协议 $\Pi_{i,v}$ 产生的联合分布，和原协议 Π 在第 i 个坐标上看到的联合分布完全一样。

这是因为模拟协议中其他坐标就是按照原来的条件分布生成的：

$$(X_j^*, Y_j^*) \sim \mu(\cdot \mid V_j = v_j).$$

因此模拟出来的数据，在分布上和原问题中条件化 $V_{\setminus i} = v_{\setminus i}$ 后的数据一致。

所以它们的互信息也相等：

$$I(X_i, Y_i; \Pi(X_1^n, Y_1^n) \mid V_i, V_{\setminus i} = v_{\setminus i}) = I(X_i, Y_i; \Pi_{i,v}(X_i, Y_i) \mid V_i).$$

因为 $\Pi_{i,v}$ 是一个 δ -error protocol for h ，根据 CIC 的定义，它的信息代价至少是

$$\text{CIC}_\delta^\mu(h).$$

因此对每个坐标 i ，都有

$$I(X_i, Y_i; \Pi \mid V) \geq \text{CIC}_\delta^\mu(h).$$

再对 $i = 1, \dots, n$ 求和，得到：

$$I(X_1^n, Y_1^n; \Pi) \geq \sum_{i=1}^n I(X_i, Y_i; \Pi \mid V) \geq n \cdot \text{CIC}_\delta^\mu(h).$$

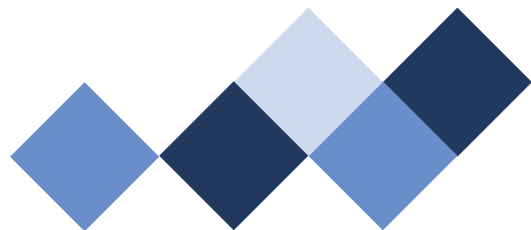
由于 Π 是任意能解决 f 的协议，所以最优协议也必须满足这个下界：

$$\text{IC}_\delta(f) \geq n \cdot \text{CIC}_\delta^\mu(h).$$

Corollary 10.3.14. Let f be the *binary inner product* $f(x, y) = \langle x, y \rangle \bmod 2$ or the *disjointness function* $f(x, y) = \mathbf{1} \{ \langle x, y \rangle > 0 \}$. Let μ be the fooling distribution in Example 10.3.12. Then

$$\text{IC}_\delta(f) \geq n \cdot \text{CIC}_\delta^\mu(h)$$

where $h(a, b) = ab$ is the product (or AND) function.





Communication complexity in estimation

普通统计估计里，估计器可以直接看到所有数据：

$$X_1, \dots, X_m.$$

然后构造估计量：

$$\hat{\theta} = \hat{\theta}(X_1, \dots, X_m).$$

但在 communication complexity in estimation 中，数据分散在 m 个机器或 agent 上：

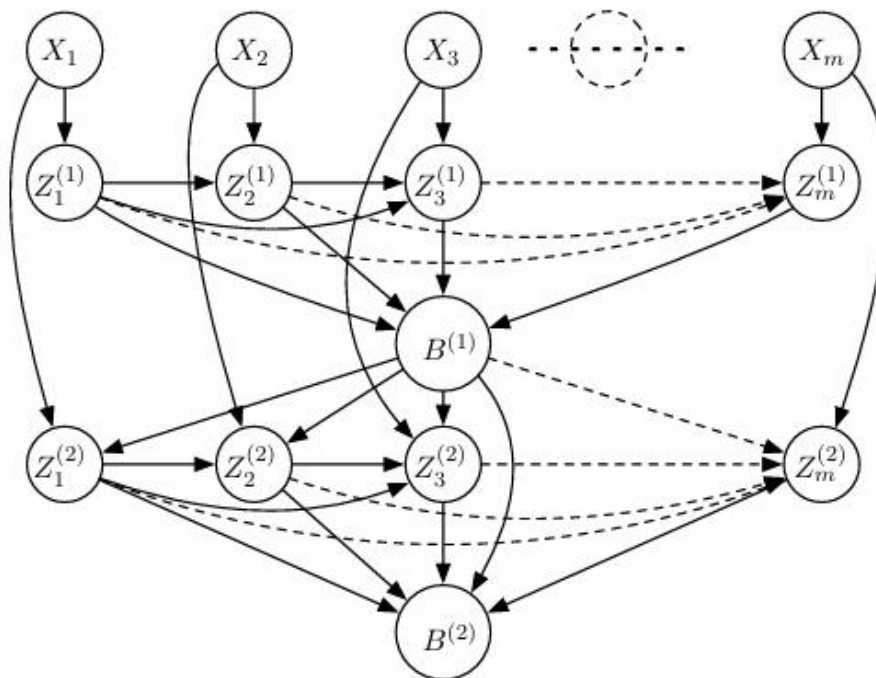
machine i has X_i .

每台机器不能直接把完整 X_i 发出去，而只能发送受限信息：

$$Z_i^{(t)}.$$

最终估计器只能根据通信 transcript / blackboard 产生估计：

$$\hat{\theta} = \hat{\theta}(B).$$



$\tau < t$, and all machines may read these public blackboards. Thus, in round t , individual i generates the communicated variable $Z_i^{(t)}$ according to the channel

$$Q_{Z_i^{(t)}}(\cdot \mid X_i, Z_{<i}^{(t)}, B^{(t-1)}) = Q_{Z_{\rightarrow i}^{(t)}}(\cdot \mid X_i, Z_{\rightarrow i}^{(t)}).$$

Here we have used the notation $Z_{<i} := (Z_1, \dots, Z_{i-1})$, and we will use $Z_{\leq i} := (Z_1, \dots, Z_i)$ and similarly for superscripts throughout. We will also use the notation $Z_{\rightarrow i}^{(t)} = (B^{(t-1)}, Z_{<i}^{(t)})$ to denote all the messages coming into communication of $Z_i^{(t)}$. Figure 10.3 illustrates two rounds of this

典型应用场景

- memory-limited estimation
- private estimation
- distributed estimation





Direct sum communication bound

$v \in \{0, 1\}^d$, and in proving our lower bounds, assume the typical Markov structure

$$V \rightarrow (X_1, \dots, X_m) \rightarrow \Pi(X_1^m),$$

where V is chosen uniformly at random from $\{-1, 1\}^d$, and $\Pi = \Pi(X_1^m)$ denotes the transcript of the entire communication, i.e., the blackboard messages

$$\Pi = (B^{(1)}, \dots, B^{(T)}),$$

(this also encodes the message order). We assume that X follows a d -dimensional product distribution, so that conditional on $V = v$ we have

$$X \stackrel{\text{iid}}{\sim} P_v = P_{v_1} \otimes P_{v_2} \otimes \dots \otimes P_{v_d}. \quad (10.4.1)$$

The generation strategy (10.4.1) guarantees that conditional on the j th coordinate $V_j = v_j$, the coordinates $X_{i,j}$ are i.i.d. and independent of $V_{\setminus j} = (V_1, \dots, V_{j-1}, V_{j+1}, \dots, V_d)$ as well as independent of $X_{i',j}$ for data points $i' \neq i$.

with the dimension. To accomplish this reduction, let $X_{\leq m,j} = (X_{i,j})_{i=1}^m$ be the j th coordinate of the data, and let $X_{\leq m,\setminus j}$ be the remaining $d-1$ coordinates across all $i = 1, \dots, m$. Then by the construction (10.4.1), we have the Markov structure

$$V_j \rightarrow X_{\leq m,j} \rightarrow \Pi(X_1^m) \leftarrow X_{\leq m,\setminus j} \leftarrow V_{\setminus j}.$$

Viewing $X_{\leq m,\setminus j}$ as extraneous randomness, we have the simpler Markovian structure

$$V_j \rightarrow X_{\leq m,j} \rightarrow \Pi, \quad (10.4.2)$$

so that we may think of the communication $\Pi = \Pi(X_{\leq m,j})$ as acting only on $X_{\leq m,j}$ while simulating $X_{\leq m,\setminus j}$. Now, define M_{-j} and M_j to be the marginal distributions over the communication protocol

假设 V in $\{-1, 1\}^d$, 可以使用 Assouad

Markov 过程

product distribution: 可以使用 direct sum

建立高维估计问题的 direct sum 下界, 把 d -维问题的误差下界归约到 d 个一维检验问题的误差下界之和





Direct sum communication bound

Π conditional on $V_j = \pm j$, the one-variable model (10.4.2). Then **Le Cam's testing equality** (Proposition 2.3.1), and the equivalence between Hellinger and variation distance (Proposition 2.2.8) imply

$$\begin{aligned} \inf_{\hat{V}} 2 \sum_{j=1}^d \mathbb{P}(\hat{V}_j(\Pi) \neq V_j) &\geq \sum_{j=1}^d (1 - \|M_{-j} - M_{+j}\|_{\text{TV}}) \geq \sum_{j=1}^d (1 - \sqrt{2} d_{\text{hel}}(M_{-j}, M_{+j})) \\ &\geq d \left(1 - \left(\frac{2}{d} \sum_{j=1}^d d_{\text{hel}}^2(M_{-j}, M_{+j}) \right)^{1/2} \right) \end{aligned}$$

by Cauchy-Schwarz. Summarizing, we have the following

Proposition 10.4.1 (Assouad's method in communication). *Let M_{+j} be the marginal distribution over Π conditional on $V_j = 1$ and M_{-j} be the marginal distribution of Π conditional on $V_j = -1$ in Markov structure (10.4.2), and assume X_i follow the product distribution (10.4.1). Then*

$$\sum_{j=1}^d \mathbb{P}(\hat{V}_j(\tau) \neq V_j) \geq \frac{d}{2} \left(1 - \sqrt{\frac{2}{d} \sum_{j=1}^d d_{\text{hel}}^2(M_{-j}, M_{+j})} \right).$$

$$\|P-Q\|_{\text{TV}} \leq \sqrt{2} d_{\text{hel}}(P,Q)$$

Cauchy

左边是 estimator 在 d 个坐标上的总错误数期望

右边由维度和 M_{-j} 与 M_{+j} 的差距控制





Definition 10.7. Let P_0, P_1 be arbitrary distributions on a space $\mathcal{X}$, let $V \in \{0, 1\}$ uniformly at random, and conditional on $V = v$, draw $X \sim P_v$. Consider the Markov chain $V \rightarrow X \rightarrow Z$. The **mutual information strong data processing constant** $\beta(P_0, P_1)$ is

$$\beta(P_0, P_1) := \sup_{X \rightarrow Z} \frac{I(V; Z)}{I(X; Z)},$$

where the supremum is taken over all conditional distributions (Markov kernels) from X to Z .

Proposition 10.4.2. Let $V \rightarrow X \rightarrow Z$, where $X \sim P_v$ conditional on $V = v$. Let P_X and $P_X(\cdot | Z)$ denote the marginal and conditional distributions on X given Z , respectively. If $|\log \frac{dP_v}{dP_{v'}}| \leq \alpha$ for all v, v' , then

$$I(V; Z) \leq 4(e^\alpha - 1)^2 \mathbb{E}_Z \left[\|P_X(\cdot | Z) - P_X\|_{TV}^2 \right] \leq 2(e^\alpha - 1)^2 I(X; Z).$$

We now give the two main results connecting mutual information and the contraction-type bounds in Definition 10.7. To provide bounds using Proposition 10.4.1, we wish to **control the Hellinger distance** between individual marginals $M_{\pm j}$, so we consider single variables in the Markov chain

$$V \rightarrow (X_1, \dots, X_m) \rightarrow \Pi,$$

where $V \in \{0, 1\}$. To state the coming theorems, we make a restriction on the data generation $V \rightarrow X$, calling distributions P_0 and P_1 **(c, β) -contractive** if

$$\beta(P_0, P_1) \leq \beta \leq 1 \quad \text{and} \quad \max \{D_\infty(P_0 \| P_1), D_\infty(P_1 \| P_0)\} \leq \log c, \quad (10.4.3)$$

where $D_\infty(\cdot \| \cdot)$ denotes the **Rényi- ∞ -divergence**. Proposition 10.4.2 shows that whenever such a c exists we certainly have $\beta(P_0, P_1) \leq 2(c - 1)^2$.

如何在通信受限的条件下，控制 M_{-j} 和 M_{+j} 的距离？

关于 V 的信息最多占关于 X 的通信信息的多少

比单纯的 $I(V; Z) \leq I(X; Z)$ 更强

$$D_\infty(P \| Q) = \log \sup_{x: q(x) > 0} \frac{p(x)}{q(x)}.$$





Theorem 10.4.4. Let $1 \leq c < \infty$ and $\beta \leq 1$. Let P_0 and P_1 be (c, β) -contractive (10.4.3) distributions on $\mathcal{X}$ and M_v , $v \in \{0, 1\}$ be the marginal distribution of the protocol Π conditional on $V = v$. Then

$$d_{\text{hel}}^2(M_0, M_1) \leq \frac{7}{2}(c+1)\beta \cdot \min \{I(X_1^m; \Pi(X_1^m) \mid V = 0), I(X_1^m; \Pi(X_1^m) \mid V = 1)\}.$$

Theorem 10.4.5. Let Π the entire communication protocol in Figure 10.3, $V \in \{-1, 1\}^d$ be uniform, and generate $X_i \stackrel{\text{iid}}{\sim} P_v$, $i = 1, \dots, m$ according to the independent coordinate distribution (10.4.1). Assume additionally that for each coordinate $j = 1, \dots, d$, the coordinate distributions $P_{\pm v_j}$ are (c, β) -contractive (10.4.3). Then for any estimator $\hat{V}$,

$$\sum_{j=1}^d \mathbb{P}(\hat{V}_j(\Pi) \neq V_j) \geq \frac{d}{2} \left(1 - \sqrt{7(c+1)\frac{\beta}{d} \cdot I(X_1, \dots, X_m; \Pi \mid V)} \right).$$

如何在通信受限的条件下，控制 M_{-j} 和 M_{+j} 的距离？

10.4.5 = Assouad + 10.4.4

先控制单个二元坐标通信在 transcript 下的 Hellinger 可区分性，再通过 Assouad 方法累加到 d 个坐标上，得到通信受限估计问题的整体错误下界

